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TCH 302 — Past Questions & Materials
by Renance

Course: TCH 302
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — TCH 302 Chapter 11 Mixing Processes Class Slides 1786594143292 0

Level 300 • Second Semester • Handout

TCH 302 — CHEMICAL ENGINEERING THERMODYNAMICS I

Chapter 11: Mixing Processes

An Outcome-Based (OBE) lecture for 300-level Chemical Engineering

Chapter Lecturer: Engr. P. F. Tarbuka

Related: Ch 4 Heat Effects (Engr. Dr. E. O. Ohimor) · Ch 14 Reaction Equilibria (Mr. F. M. Osunde)

OUTCOME-BASED EDUCATION

Learning Outcomes

By the end of this chapter, you should be able to:

LO1 — Define a mixing process and the property change of mixing. Understand

LO2 — Explain partial molar properties and the summability / Gibbs-Duhem relations. Understand

LO3 — Compute ΔV , ΔH , ΔS , ΔG for ideal mixing. Apply

LO4 — Use excess properties and heats of mixing for real solutions. Apply / Analyse

LO5 — Connect mixing to Heat Effects (Ch 4) and Reaction Equilibria (Ch 14). Analyse

LO6 — Interpret industrial mixing operations, especially in oil & gas. Evaluate

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INTRODUCTION

What Is a Mixing Process?

A mixing process brings two or more streams together to form one uniform solution. Why engineers care

Analogy — a cup of tea. Stir sugar into hot water and three things happen at Almost nothing in a plant is pure:

once:

- Crude oil — a mixture
- The sugar spreads out → disorder (entropy) increases.
- Natural gas — a mixture
- The cup warms or cools slightly → an energy / heat effect.
- Drug formulations — mixtures
- The final volume $\neq$ water + sugar volume → a volume effect.
- Treated water — a mixture

Those three effects — volume, energy, entropy — are the whole of mixing To size a blend tank or predict a temperature

thermodynamics. rise, you must know how properties change on mixing.

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CORE DEFINITION

Property Change of Mixing

How much a property changed because mixing happened:

$$\Delta M = M - \sum x_i M_i$$

$$M = M = \sum x_i M_i \quad (M = M - M = 0)$$

M stands for any molar property — V, U, H, S or G — giving the four quantities engineers want:

$$\Delta V \quad \Delta H \quad \Delta S \quad \Delta G$$

Volume change Heat of mixing Entropy change Gibbs change

blend-tank sizing, density does the tank heat or cool? the driving force for mixing does mixing happen on its own?

Signs to memorise: $\Delta H < 0$ warms up (acid into water) · $\Delta H > 0$ cools · $\Delta S > 0$ almost always · $\Delta G < 0$ spontaneous

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THE HEART OF MIXING

Partial Molar Properties

A species behaves differently inside a mixture than when pure.

Analogy — a crowded bus. One extra passenger takes up more or less effective space depending on how full the bus already is.

Partial molar property M_i = the change in the total property per mole of i added, holding T, P and all other amounts fixed.

Summability: $M = \sum x_i M_i$

Gibbs-Duhem: $\sum x_i dM_i = 0$ (const T, P)

See-saw analogy: if one partial molar value moves, the other must adjust to keep balance.

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THE SIMPLE CASE

Ideal Solutions & Ideal Mixing

Ideal solution: molecules treat each other exactly like their own kind — no special heat or volume effect.

Analogy — mixing red and blue marbles of identical size and weight: only the arrangement (entropy)

changes.

$$\Delta G(\text{id}) = RT \sum x = \ln x = < 0$$

$$\Delta S(\text{id}) = -R \sum x = \ln x = > 0$$

$$\Delta H(\text{id}) = 0 \quad \Delta V(\text{id}) = 0$$

All the action is entropy — mixing

is most favourable at $x = 0.5$.

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WHEN REALITY BITES

Real Solutions & Excess Properties

Excess property = deviation of the real solution from an ideal one:

$$M = M - M(\text{id})$$

Two shortcuts: $H = \Delta H(\text{mix})$ and $V = \Delta V(\text{mix})$.

Activity coefficient γ_j scores the non-ideality:

- $\gamma = 1$ ideal · $\gamma > 1$ molecules 'dislike' (endothermic) · $\gamma < 1$ they 'like' (exothermic)

$$G = RT \sum x = \ln \gamma$$

Analogy — a group project: cooperative teammates ($\gamma < 1$) release energy; clashing ones ($\gamma > 1$) waste it.

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WHAT YOU FEEL IN THE PLANT

Heats of Mixing

The heat of mixing shows up as a temperature change.

Safety rule — 'add acid to water, never water to acid': the large water mass absorbs the exothermic heat so the temperature rise stays small.

Adiabatic mixer energy balance:

$$\Delta T = -\Delta H(\text{mix}) / (m \cdot c_p)$$

Enthalpy-concentration (H-x) charts turn mixing and heat problems into simple lever-rule constructions.

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WORKED EXAMPLE

Entropy & Gibbs of Ideal-Gas Mixing

Mix 2 mol N₂ with 3 mol O₂ at 300 K, same P → 5 mol ideal-gas mixture. Find ΔS and ΔG of mixing.

Step 1 — Mole fractions: $y(\text{N}_2) = 2/5 = 0.4$, $y(\text{O}_2) = 3/5 = 0.6$.

Step 2 — $\sum y_j \ln y_j = 0.4 \ln 0.4 + 0.6 \ln 0.6 = -0.673$.

Step 3 — Per mole: $\Delta S(\text{id}) = -R(-0.673) = +5.60 \text{ J/mol}\cdot\text{K} \rightarrow \text{for 5 mol} = 28.0 \text{ J/K}$.

Step 4 — Per mole: $\Delta G(\text{id}) = RT(-0.673) = 8.314 \times 300 \times (-0.673) = -1678 \text{ J/mol} \rightarrow \text{for 5 mol} = -8.39 \text{ kJ}$.

Conclusion: $\Delta G < 0 \rightarrow$ gases mix spontaneously, driven entirely by the entropy rise. Un-mixing (air separation) costs work.

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BRIDGE — CHAPTER 4

Link to Heat Effects

Enthalpy is a state function, so any process is the sum of standard 'legs':

$$\Delta H(\text{overall}) = \Delta H(\text{sensible}) + \Delta H(\text{latent}) + \Delta H(\text{mixing}) + \Delta H(\text{reaction})$$

Analogy — a journey by legs: total distance north is the sum of each leg, whatever road you take.

Chapter 11 supplies the mixing leg ($\Delta H(\text{mix}) = H_J$); Chapter 4 supplies the sensible leg ($\int c_p dT$), the latent leg, and the reaction leg.

Same accounting framework — different legs. That is why the two chapters share one energy balance.

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BRIDGE — CHAPTER 14

Link to Chemical Reaction Equilibria

Reactions happen inside mixtures. Equilibrium is set by activities — and activities come from Chapter 11.

$$\Delta G^\circ = -RT \ln K \quad K = \prod (\hat{a})^{\nu} = \hat{a} = \gamma = x =$$

The chemical potential driving a reaction is just the partial molar Gibbs energy:

$$\bar{\mu} = M_J(G) = G_J + RT \ln \hat{a}_J.$$

If you pretend $\gamma = 1$ when it is not, the effective concentration — and the predicted conversion — can be off by tens of percent.

Get the mixing thermodynamics wrong and the equilibrium prediction is wrong.

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WHERE IT IS USED

Industrial Applications

Oil & Gas (primary) Water Treatment

- Crude & product blending to spec (density, sulfur, ...)
- Rapid-mix coagulation (alum, ferric ...)

octane)

chloride)

- Amine gas sweetening — heat of absorption/mixing • Neutralising acidic/alkaline effluents —
- Glycol dehydration; EOR surfactant/CO₂ mixing heat release
- Safe acid/chemical dosing (temperature control)

Pharmaceuticals Food & Process Industries

- Blending actives, solvents & excipients powders
- Dissolving sugars/salts, reconstituting
- Paints, detergents, polymer solutions —
- Heats of solution affect stability tank & agitator design & content uniformity

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PULLING IT TOGETHER

Chapter Summary & Key Equations

Mixing changes properties: $\Delta M = M - \sum x_i M_i$

M_i = Key equations

- Partial molar M describe species inside a $\Delta M = M - \sum x_i M_i =$ mixture; summability + Gibbs–Duhem link

them. $M = \sum x_i M_i$

M_i (summability)

- Ideal mixing is pure entropy: $\Delta H = \Delta V = 0$, $\sum x_i M_i$

$dM_i = 0$ (Gibbs–Duhem)

$\Delta S > 0$, $\Delta G < 0$. $\Delta G(\text{id}) = RT \sum x_j \ln x_j$

- Real solutions: excess properties + activity $\Delta S(\text{id}) = -R \sum x_j \ln x_j$ coefficients ($H_j = \Delta H(\text{mix})$).

$G_j = RT \sum x_j \ln \gamma_j$

- Heat of mixing → temperature change: $\Delta T =$

$-\Delta H(\text{mix}) / (m c_p)$. $\Delta T = -\Delta H(\text{mix}) / (m c_p)$

- Ch 4 link: mixing is one leg of enthalpy $K = \prod (\gamma_j x_j)^{\nu_j}$, $\Delta G^\circ = -RT \ln K$ accounting.

- Ch 14 link: activities set the equilibrium constant.

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Review & Assessment

Quick questions before you leave:

1. Why is $\Delta S(\text{id})$ always positive?
2. A blend heats up on mixing — what is the sign of ΔH , and is this ideal?
3. How does an activity coefficient enter the equilibrium constant of Chapter 14?

4. Name one oil-and-gas operation dominated by the heat of mixing.

Full worked problems: see the Chapter 11 tutorial sheet (10 graded problems).